Exploring Mobile USB port Communication with Analogue Device METAL DETECTOR APPLICATION

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Abstract

The Mobile Metal Detector Application revolutionizes metal detection capabilities by seamlessly integrating smartphone functionality with external metal detectors, enhancing security, enabling archaeological discoveries, and facilitating outdoor activities.

The application's innovative design harnesses the power of the smartphone's USB port, enabling seamless communication with external detectors for enhanced performance and versatility. This documentation provides a comprehensive overview of the development process, focusing on system architecture, design considerations, user interface details, and testing strategies. By chronicling the meticulous planning, implementation, and evaluation stages, this document offers insights into the creation of a cutting-edge mobile tool that leverages the full potential of smartphone technology and external hardware integration.

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CHAPTER 1 SOFTWARE PROJECT MANAGEMENT PLAN (SPMP)

1. Introduction

1.1 Project Overview

Our project revolves around exploring the utilization of the USB port for communication with an analog metal detector, enhancing the process of metal detection through innovative mobile software integration. This mobile application, will be developed using Android Studio and Kotlin, aims to seamlessly integrate with compatible metal detectors. By tracking key parameters such as position, intensity, and time, our application will offer an efficient approach to metal detection, enabling users to gather valuable insights for enhanced metal detection processes.

While wireless communication may seem convenient, it often introduces significant noise in data reception. To ensure accurate data analysis and reliable performance on mobile devices, we've opted for a wired connection method. This decision enables us to maintain data integrity and streamline the detection process effectively.

1.1.1 Motivation:

The primary goal of this project is to explore the connectivity of the USB port for analog external devices, particularly in the context of metal detection. Our motivation stems from the desire to create dynamic and efficient tools and techniques for metal detection, leveraging everyday devices such as smartphones by connecting them to external detectors.

The necessity for this application arises from the limitations of present conventional stationary metal detectors, which often pose challenges in usability and accessibility. Our application aims to surpass the functionalities of these traditional tools while providing users with the convenience of utilizing their smartphones for metal detection tasks. By bridging the gap between mobile technology and metal detection, we seek to offer users a fluid and diverse experience while meeting the needs and applications of this field effectively.

1.1.2 Application:

This Metal detection mobile application has a lot of utilities in a wide range of scenarios. It can serve as the valuable instrument for security personnel, event organizers and law enforcement, elevating security through real time detection of concealed metal objects
Furthermore, it can fulfill the needs of outdoor enthusiasts, archaeologists or various metal-related activities, enhancing their exploration and finding experiences

1.1.3 Existing systems:

In discussing existing systems, it's crucial to highlight the capabilities of USB connectivity, particularly in the context of Android devices. This includes an exploration of the existing support for USB connectivity at the Android level, along

with available APIs and frameworks that facilitate communication between external devices and smartphones.

One of the key distinctions between conventional metal detectors and our application lies in its ability to seamlessly integrate with external metal detectors. This collaborative approach leverages the strengths of both smartphones and external devices, resulting in a more versatile, portable, and user-friendly solution. The application's distinguishing features include real-time alerting and storage capabilities, as well as robust management functionalities for detection information. These aspects set it apart from existing solutions in terms of innovation and functionality.

1.2 Project Deliverable

Following are the the key deliverable of the "Metal Detector Mobile Application" project:
1. Android Mobile App:
A Mobile application with an intuitive design that makes it easier to interact with external metal detectors and records raw data for further analysis.

2. Integration with External Metal Detectors:
For communication and data transfer between the mobile application and external detector device we establish seamless protocol ensuring an efficient collaboration.
3. Project Documentation:
By the extensive documentation a detailed reference for stakeholders is provided that covers the project's goals, design specifications, implementation specifics, and testing procedures.
4. Testing and Quality Assurance:
To confirm the application's accuracy, functionality, and dependability, guaranteeing a stable and reliable performance we are setting strict testing processes.

1.3 Project Organization

1.3.1 Software Process Model

The development of the Mobile Application will follow an iterative and incremental software process model. This approach allows for continuous improvement based on user feedback and evolving requirements. It encourages early user involvement and ensures the development of a robust and user-friendly final product.

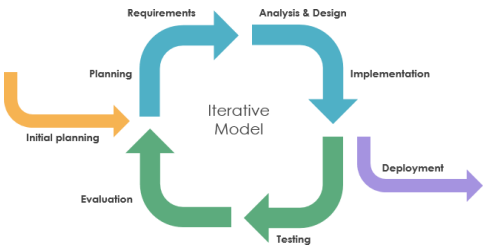


Figure 1.1: Software Process Model (Iterative Model)

1.3.2 Roles and Responsibilities

1. App Developer:
   - Designs and implements the mobile app's user interface and functionality.
   - Collaborates on connectivity with external metal detectors.

2. UI/UX Designer:
   - Develops intuitive and user-friendly interface designs.
   - Ensures a seamless and visually appealing user experience.

3. Hardware Integration:
   - Integrates the app with metal detectors using wired connections.
   - Ensures compatibility with various metal detectors.

4. Data Analyst:
   - Develops algorithms for processing raw detector data into analyzable digital data.
   - Collaborates with the app development team for seamless integration.

5. QA Tester:
   - Conducts thorough testing for functionality, accuracy, and reliability.
   - Reports and addresses any identified issues.

6. Project Manager:
   - Oversees project progress and milestones.
   - Facilitates communication and team collaboration.

7. Documentation Specialist:
   - Compiles and maintains project documentation.
   - Ensures comprehensive and accessible records.

1.3.3 Tools and Techniques

1. Android Studio IDE:
   - Primary development environment for creating the Android application.
   - Offers a comprehensive set of tools for app design, coding, debugging, and testing.

2. Kotlin Programming Language:
   - Chosen language for application development, known for conciseness and enhanced code readability.
   - Compatible with Android Studio, facilitating efficient development.

3. Firebase Real time Database:
   - Utilized for data storage and management, ensuring real-time accessibility for users.
   - Enables seamless integration with the mobile application for efficient data handling.

4. Wire Communication Protocols:
  - Wired connections technology is employed for effective communication with external metal detectors.
  - Ensures reliable data transfer and synchronization between the app and detectors.

5. Algorithms for Data Processing:
  - Developed algorithms for processing raw data generated by the metal detector.
  - Converts raw data into analyzable digital data for effective interpretation.

6. Testing Frameworks:
  - Various test frameworks are used for comprehensive quality assurance.
  - guarantees the functionality, accuracy and reliability of the mobile application.

1.4. Project Management Plan

1.4.1 Tasks

The project management plan for the "Metal Detector Mobile App" includes several key tasks:

1.4.2 Description

1.4.2.1 Requirements and Analysis Phase

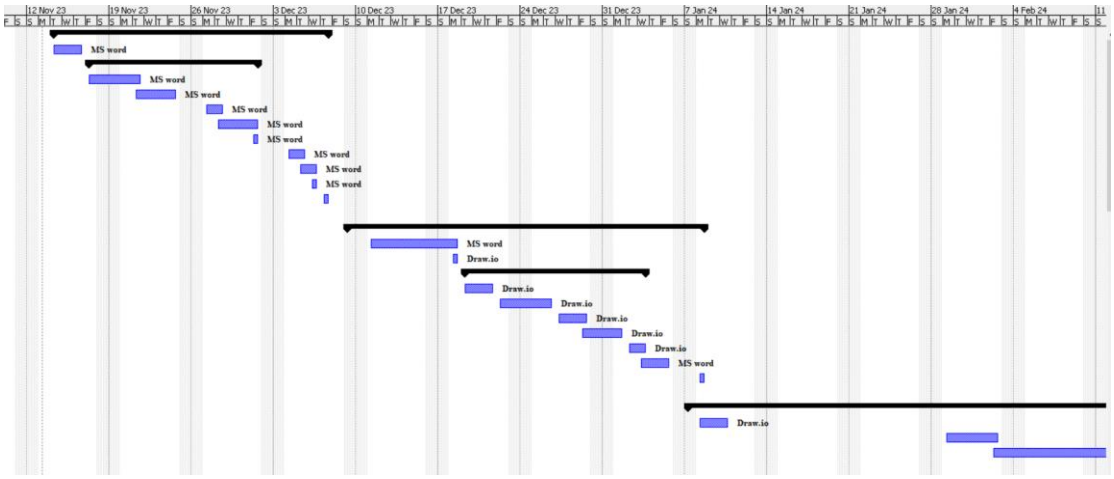
- Objective: Identify and define the functional and non-functional requirements of a mobile application.
- Activities:
  - Work with stakeholders to gather requirements.
  - Analyze collected data to understand user needs and system constraints.
  - Based on requirements defines the scope of Project.

1.4.2.2 Design phase

- Goal: Develop a comprehensive mobile application design, outline the user interface and system architecture.
- Activities:
  - Use collected requests to create a user-friendly interface.
  - Define the overall architecture of the application with regard to integration with external metal detectors.
  - Prepare detailed project documentation for implementation.

Gantt Chart:

		Name	Duration	Start	Finish	Resource Names
1		**Planning**	18 days	**11/14/23 8:00 AM**	**12/7/23 5:00 PM**	
2		define purpose	3 days	11/14/23 8:00 AM	11/16/23 5:00 PM	MS word
3		**define scope**	11 days	**11/17/23 8:00 AM**	**12/1/23 5:00 PM**	
4		Stakeholders analysis	3 days	11/17/23 8:00 AM	11/21/23 5:00 PM	MS word
5		define major functions	4 days	11/21/23 8:00 AM	11/24/23 5:00 PM	MS word
6		define inputs/outputs	2 days	11/25/23 8:00 AM	11/28/23 5:00 PM	MS word
7		define objectives	4 days	11/28/23 8:00 AM	12/1/23 5:00 PM	MS word
8		define constraints	1 day	12/1/23 8:00 AM	12/1/23 5:00 PM	MS word
9		define process model	2 days	12/2/23 8:00 AM	12/5/23 5:00 PM	MS word
10		define tools and techniques	2 days	12/5/23 8:00 AM	12/6/23 5:00 PM	MS word
11		define project deliverables	1 day	12/6/23 8:00 AM	12/6/23 5:00 PM	MS word
12		Milestone: Completed	1 day	12/7/23 8:00 AM	12/7/23 5:00 PM	
13		**Analysis**	21 days	**12/9/23 8:00 AM**	**1/8/24 5:00 PM**	
14		Describing use cases	6 days	12/9/23 8:00 AM	12/18/23 5:00 PM	MS word
15		develop use case diagram	1 day	12/18/23 8:00 AM	12/18/23 5:00 PM	Draw.io
16		**develop analysis models**	12 days	**12/19/23 8:00 AM**	**1/3/24 5:00 PM**	
17		develop domain model	3 days	12/19/23 8:00 AM	12/21/23 5:00 PM	Draw.io
18		develop state transition diagran	3 days	12/22/23 8:00 AM	12/26/23 5:00 PM	Draw.io
19		develop activity diagram	3 days	12/27/23 8:00 AM	12/29/23 5:00 PM	Draw.io
20		develop system sequence diagra	2 days	12/29/23 8:00 AM	1/1/24 5:00 PM	Draw.io
21		develop data model	2 days	1/2/24 8:00 AM	1/3/24 5:00 PM	Draw.io
22		develop SRS	3 days	1/3/24 8:00 AM	1/5/24 5:00 PM	MS word
23		review SRS	1 day	1/6/24 8:00 AM	1/8/24 5:00 PM	
24		**Design**	35 days	**1/7/24 8:00 AM**	**2/23/24 5:00 PM**	
25		Develop architecture diagram	3 days	1/7/24 8:00 AM	1/10/24 5:00 PM	Draw.io
26		database design	5 days	1/29/24 8:00 AM	2/2/24 5:00 PM	
27		User interface design	14 days	2/2/24 8:00 AM	2/21/24 5:00 PM	Figma;Marvel
28		Review of design	2 days	2/21/24 8:00 AM	2/22/24 5:00 PM	
29		Milestone:completion	1 day	2/23/24 8:00 AM	2/23/24 5:00 PM	



1.5 Summary

In summary, the project management plan includes two key phases – the requirements and analysis phase, focused on understanding user needs and defining the scope of the project, and the design phase, where the application's user interface and system architecture are carefully planned. These phases lay the foundation for the successful development and implementation of the “Exploring Mobile USB port Communication with Analogue Device (Metal Detector Application) ”.

CHAPTER 2 SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Introduction

This chapter of the document specifies the general factors that affects the product and its requirements, providing a background for the requirements of the software. It also describes the summary of the functions that the software will perform. It also describes the user capabilities and their interest.

2.1 Product Overview

2.1.1 Stakeholders

The success of the "Mobile-based Metal Detector Scanner Application" is intricately tied to the satisfaction of various stakeholders who have a vested interest in the application's development and performance.

- End Users:

- Individuals utilizing the mobile application for metal detection purposes, including security personnel, construction workers, and archaeologists.

- Developers:

- The team responsible for the design, coding, and testing of the application.

- Project Managers:

- Overseeing the project's progress, ensuring it aligns with the defined objectives and meets stakeholders' expectations.

-Regulatory Authorities:

- Regulatory Compliance Oversight Entities:

Organizations responsible for ensuring obedience to regulations dealing both data privacy and metal detection practices.

2.1.2 Major Functions

To meet the user needs the "Mobile-based Metal Detector Scanner Application" is designed in a way to have several important functions.

- Real-time Metal Detection:

-For metal detection activities provide a real time tracking which includes time, location and intensity of the detection.

- External Detector Integration:

-For efficient Data Collection, application has ordered integration with external device through wire.

- User Interface Design:

-For Better user experience user friendly interface is developed.

2.1.3 Major Inputs and Outputs

It is important to understand the inputs and outputs of the application for effective functionality.

- Major Inputs:

- Primary data from external metal detector device , including information on metal detection events.

-Major Outputs:

-
Displaying the result of detection after analyzing the data sent from external device, also the location where any metal found.
Following segment provides a broad overview of the mobile application , identifying key stakeholders, major functions, and the inputs and outputs that defines the operations of "Metal Detector Mobile Application."
Definitions, Acronyms and Abbreviations

Acronyms	Definitions
UC	Use Case
Users	Tutor, Student
Admin	Administrator

Table 2. 1 Acronyms and Definitions

2.2 Specific Requirements

2.2.1 External Interface Requirements

2.2.1.1 User Interfaces

- Intuitive Design:
- To provide users with different levels of technological proficiency the User interface should be intuitive.
- Display tracking information, including latitude, longitude , time, and detection intensity.

2.2.1.2 Software Interfaces

-External Detector Integration:
- Uninterrupted connectivity with external metal detector device through wire.
- For data transfer protocol well organized software integration.

2.2.1.3 Hardware Interface

- Compatibility:
- For guaranteeing optimal performance on different screen sized and resolutions app will be Compatible with wide range of Android devices.

2.2.1.4 Communication Protocols

- Wired Communication:
- For connection with External metal detector (wired) there should be established a reliable communication.
-

For correct tracking of metals there should be Data Synchronization.

2.3 Software Product Features

- Real-time Tracking:
- Displaying metal detection in real time.

- External Detector Integration:
- For flawless data collection there will be efficient integration with external metal detector device.

- User-friendly Interface:
- For a positive user experience intuitive design.

2.4 Software System Attributes

2.4.1 Reliability

- Error Handling:
- For system stability implementation of robust error handling mechanisms .

2.4.2 Availability

- Continuous Availability.
- During metal detection activities the application should maintain availability.

2.4.3 Security

- Data Encryption:
- To ensure the security of sensitive information implementation of data encryption.

2.4.4 Maintainability

- Modular Design:
- Develop the application with a modular structure for ease of maintenance and future enhancements.

2.4.5 Portability

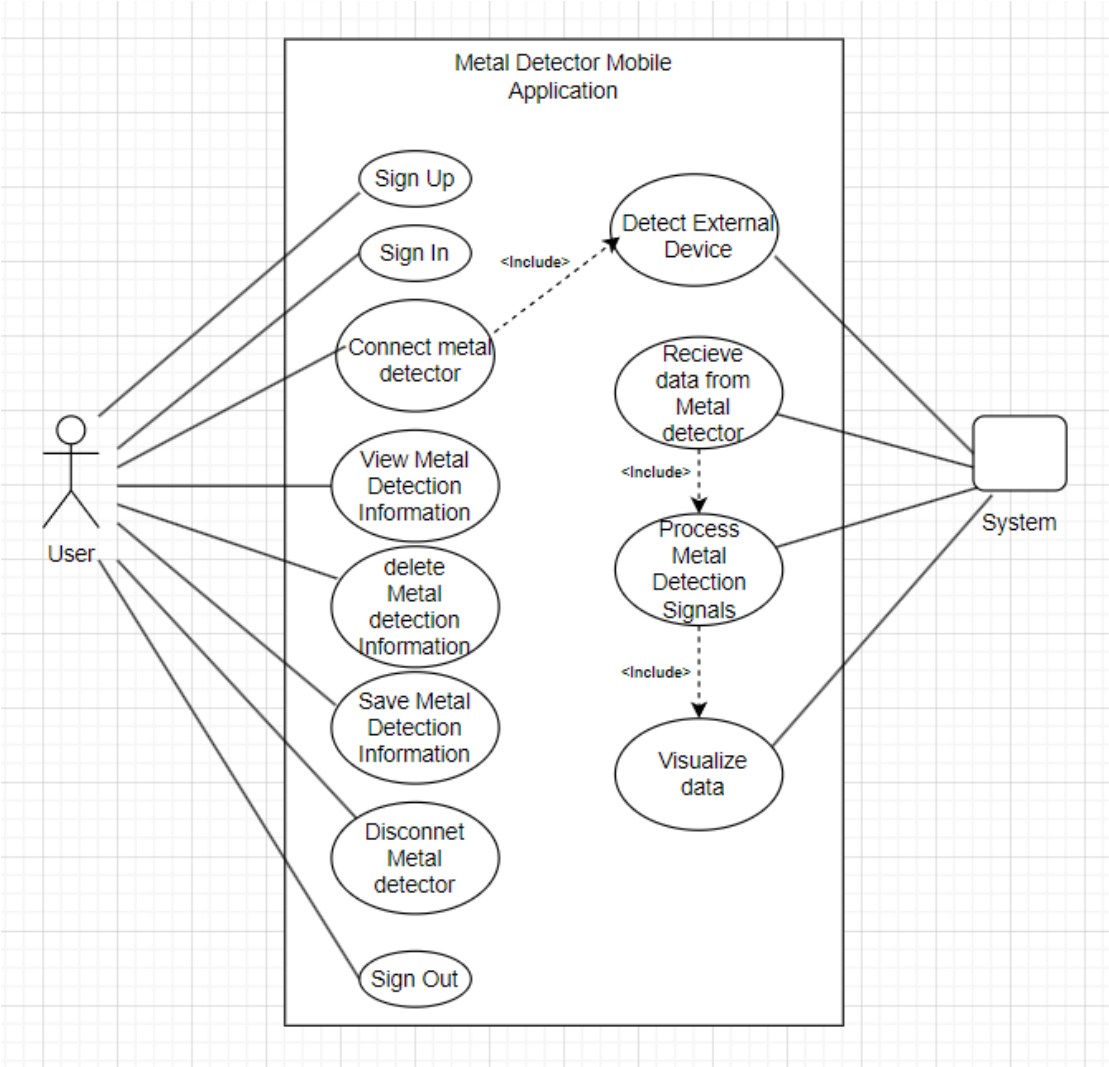
- Device Compatibility:
- Ensure portability across a variety of Android devices to maximize accessibility.

2.4.6 Performance

- Optimized Performance:
- Optimize application performance for efficient real-time tracking and data processing.

This section outlines the specific requirements related to external interfaces, software product features, and various software system attributes essential for the successful development and operation of the "Mobile-based Metal Detector Scanner Application."

2.5 Use Case Diagram :



2.6 List of Use Cases

1. Sign Up
2. Sign In
3. Connect Metal detector
4. View Metal detection Information
5. Save the Metal detection information
6. Disconnect the Device
7. Delete Saved signal information
8. Logout
9. Receive Data from Metal detector
10. Detect External Device
11. Process Metal detection signals
12. Visualize data

1. Sign Up

ID	UC-1
Name	Create Account
Primary Actors	User
Pre Condition	The user is not registered in the system.
Post Condition	The user successfully creates a new account.
Main Scenario	1. User initiates the sign-up process. 2. User provides necessary information (username, email, password). 3. System validates the provided information. 4. If validation is successful, the system creates a new account. 5. User receives confirmation of successful account creation.
Alternative Scenario	If the provided email is already associated with an existing account, the system notifies the user to choose a different email.

2. Sign In

ID	UC-2
Name	Login
Primary Actors	User
Pre Condition	The user has a registered account.
Post Condition	The user successfully logs into the system.
Main Scenario	1. User enters login credentials (username and password). 2. System validates the provided credentials. 3. If validation is successful, the user is granted access to the system. 4. User gains access to app features and functionalities.
Alternative Scenario	User enters incorrect credentials (wrong username or password). System prompts the user that the login credentials are incorrect.

	User exceeds the maximum allowed login attempts, and the system locks the account temporarily

3.Connect Metal Detector

ID	UC-3
Name	Detect and Connect Metal detector
Primary Actors	User
Pre Condition	The mobile application is installed and running. The metal detector is within range and available for connection
Post Condition	Successful connection established between the mobile application and the metal detector.
Main Scenario	1. User initiates the connection process from the application 2. The App asks for USB permission. 3. The application scans for available metal detectors connected to Mobile. 4. The application establishes a connection with the selected metal detector through a wired connection.
Alternative Scenario	If no metal detectors are found, the application notifies the user and suggests checking the metal detector's availability or ensuring the proper connection

4. View Saved Information

ID	UC-5
Name	View Information
Primary Actors	User
Pre Condition	The mobile application is installed, and there is previously saved metal detection information.
Post Condition	User successfully views the saved metal detection information.
Main Scenario	1. User accesses the "View Saved Information" feature in the application. 2. A list of previously saved metal detection information is displayed. 3. User selects a specific entry to view detailed information. 4. The application presents the selected metal detection entry, including date,

	time, and detection intensity
Alternative Scenario	If there is no saved information, the application notifies the user and suggests performing a metal detection scan to generate new data.

5. Save Processed Signal Information

ID	**UC-6**
Name	Save Processed Signal Information
Primary Actors	User
Pre Condition	Metal detection information is processed.
Post Condition	Processed information is saved in the system..
Main Scenario	1. User processes metal detection information. 2. System prompts user to save the processed information. 3. User provides a name for the detected object. 4. System saves the processed information with the provided name.
Alternative Scenario	User tries to save information without detecting any metal. System prompts the user that there is no metal detection information to save

6.Disconnect the Metal detector

ID	**UC-7**
Name	Disconnect the connected device
Primary Actors	User
Pre Condition	The app is in an active state and device is connected.
Post Condition	App settings are reset or session data is cleared.
Main Scenario	1. User initiates the reset process by clicking on disconnect button. 2. System clears session data or resets app settings.

Alternative Scenario	User selects the disconnect option accidentally. System prompts the user for confirmation before proceeding with the reset. User cancels the reset operation

7. Delete Saved Signal Information

ID	**UC-9**
Name	Delete Saved Signal
Primary Actors	User
Pre Condition	There are saved entries in the system.
Post Condition	A selected entry is successfully deleted.
Main Scenario	1. User selects a saved metal detection entry to delete. 2. System removes the selected entry from the list.
Alternative Scenario	User tries to delete information for a non-existing metal detection entry. System notifies the user that the selected entry for deletion does not exist.

8.Logout

ID	**UC-10**
Name	User Logout
Primary Actors	User
Pre Condition	The user is logged into the system.
Post Condition	The user successfully logs out.
Main Scenario	1. User initiates the logout process. 2. System terminates the user's session. 3. User is securely logged out of the system.
Alternative Scenario	User attempts to logout while in the middle of a critical operation. System prompts the user to confirm the logout, warning that unsaved changes may be lost. User cancels the logout operation.

9.Receive datafrom metal detector

ID	UC-11
Name	Receive data from external device
Primary Actors	System
Pre Condition	External device is powered on and actively looking for metal.
Post Condition	Data from metal detector is received and ready for processing.
Main Scenario	External metal detection device detects metal. Metal detection data is transmitted to the mobile application. The mobile application receives and buffers the data for processing.
Alternative Scenario	If there is a transmission error or loss of data: The mobile application attempts to re-establish communication with the external device. If unsuccessful, an error message is displayed, and the user may attempt to troubleshoot the connection.

10.Detect External Device

ID	UC-11
Name	Detect External Device
Primary Actors	System
Pre Condition	The mobile application is running and ready to detect the external device.
Post Condition	The external device is detected and ready for communication.
Main Scenario	The mobile application scans for available external metal detection devices. Upon detection, the device is identified and listed within the application interface.
Alternative Scenario	If no external devices are detected: The mobile application displays a message indicating the absence of available devices. User may check for connectivity issues or ensure the device is powered on before

	rescanning.

11.Process metal detection Data

ID	UC-14
Name	Processing the data
Primary Actors	System
Pre Condition	Metal detection data is received and buffered.
Post Condition	Processed data is ready for visualization and analysis.
Main Scenario	The mobile application receives metal detection data. Data processing algorithms analyze the received data. Processed data is formatted and prepared for visualization.
Alternative Scenario	If there is an error in processing the data: The mobile application may display an error message indicating the issue. User may attempt to troubleshoot or reprocess the data.

12.Visualize metal detection Data

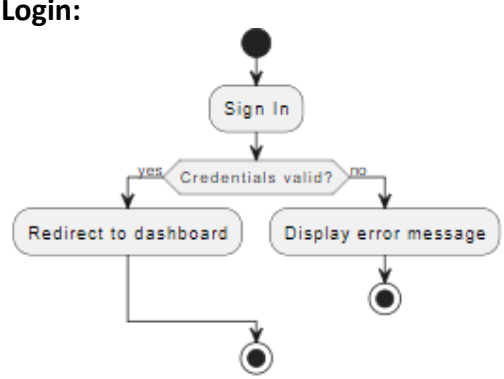
ID	UC-11
Name	Visualize Data
Primary Actors	System
Pre Condition	Processed data is available for visualization.
Post Condition	Visual representation of metal detection data is displayed to the user.
Main Scenario	Metal detection data is displayed in the mobile application. The application retrieves the processed data.
Alternative Scenario	Metal detection data is displayed in the mobile application. The application retrieves the processed data.

These detailed use case scenarios provide a comprehensive understanding of the various interactions and functionalities within the "Mobile-based Metal Detector Scanner Application."

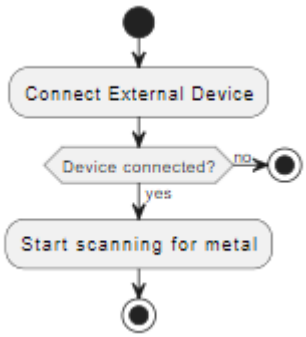
2.7 Activity Diagram:

Sign up:

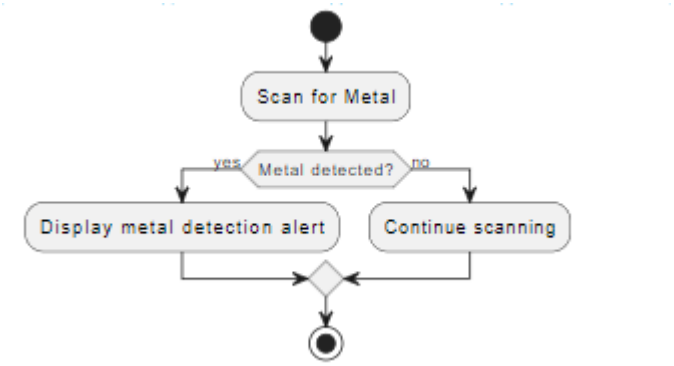
Login:



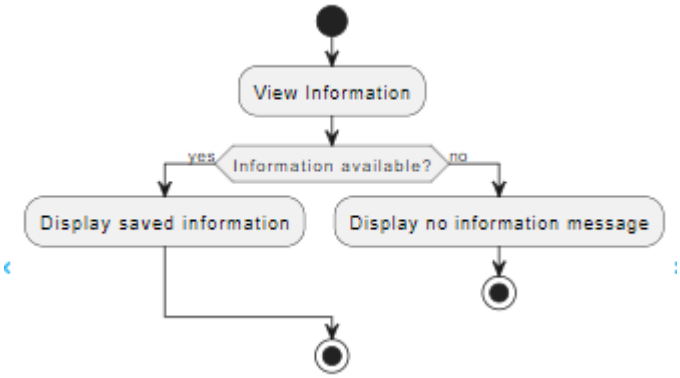
Connect External device :



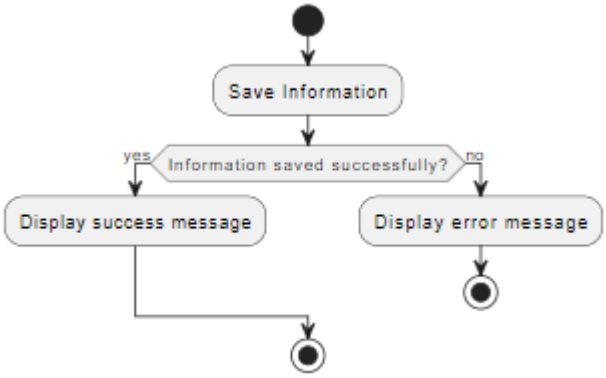
Scan For Metal:



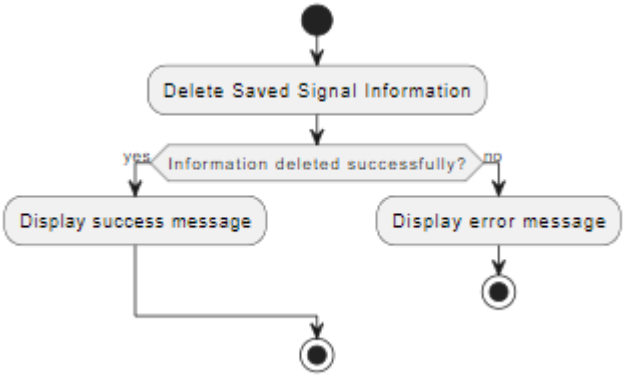
View Metal detection Information:



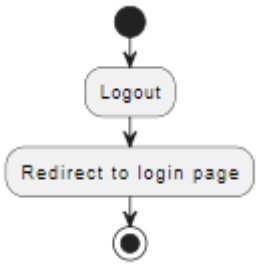
Save Metal detection Information:



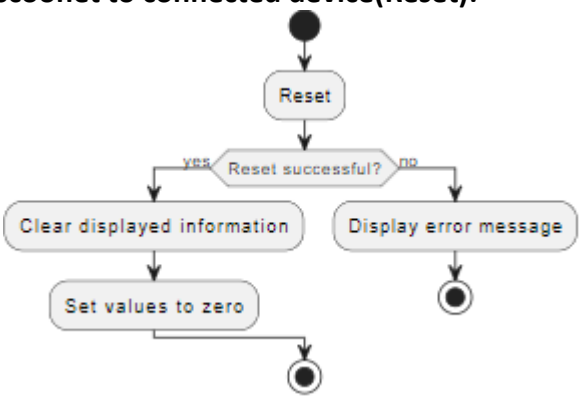
Delete saved Metal detection Information:



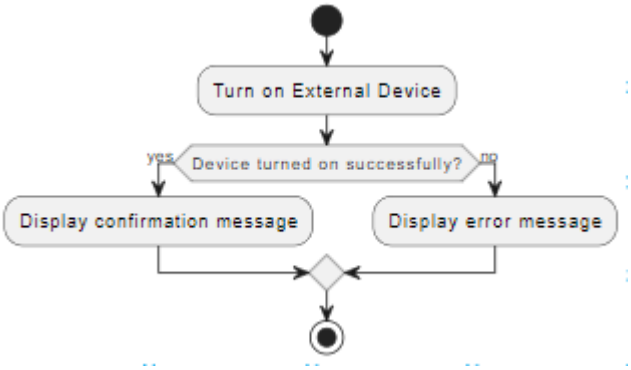
Logout:



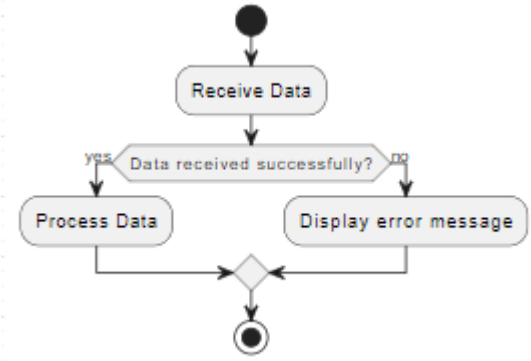
Discoonet to connected device(Reset):



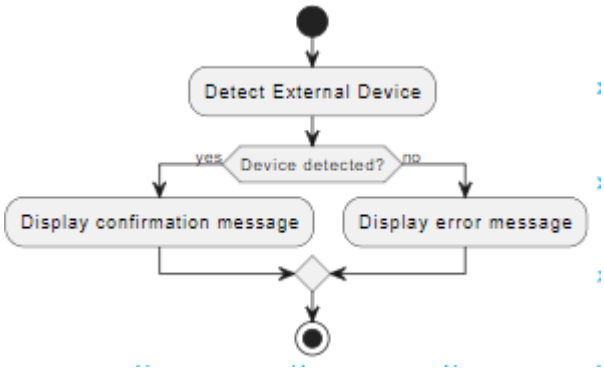
Turn On External Device:



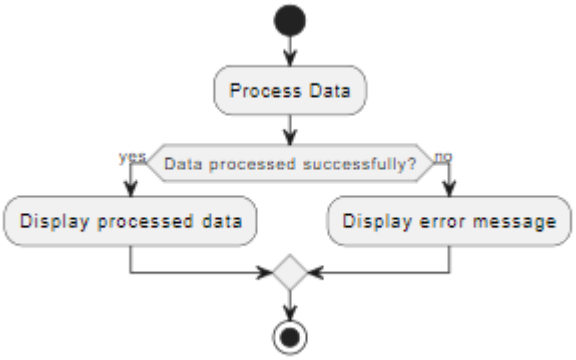
Receive Metal detection Data:



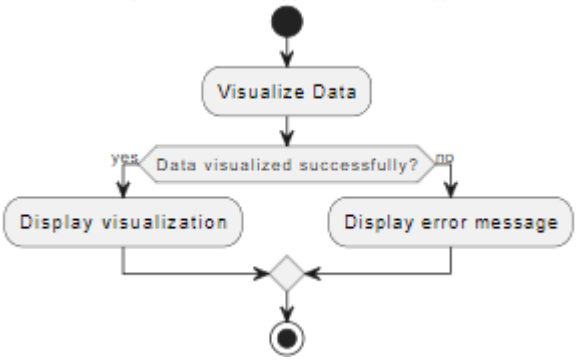
Detect External Device:



Process Metal detection Data:

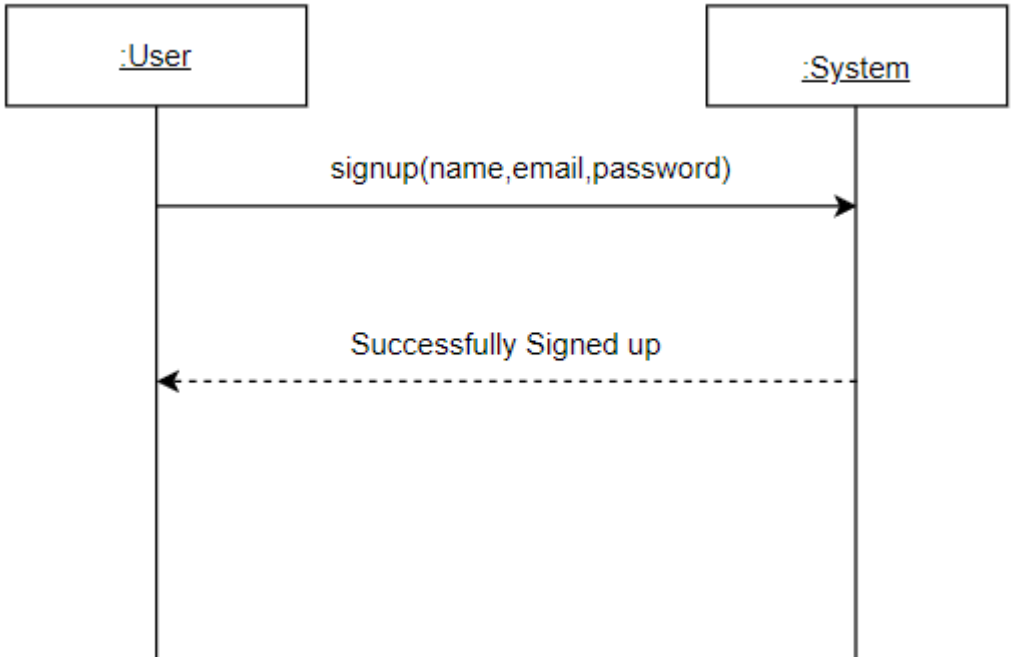


Visualize Metal detection Data:

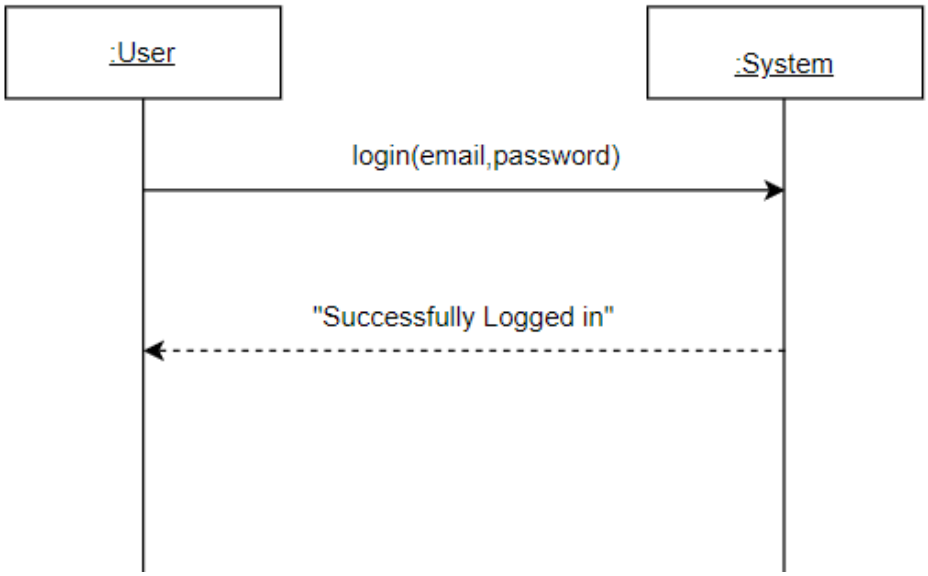


2.8 SYSTEM SEQUENCE DIAGRAM

SIGN UP:

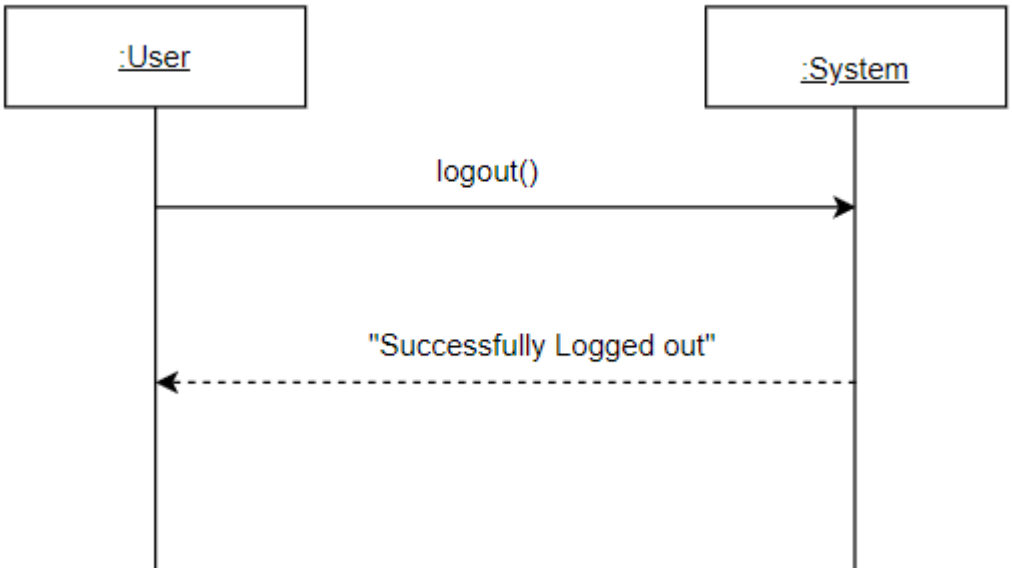


Login:

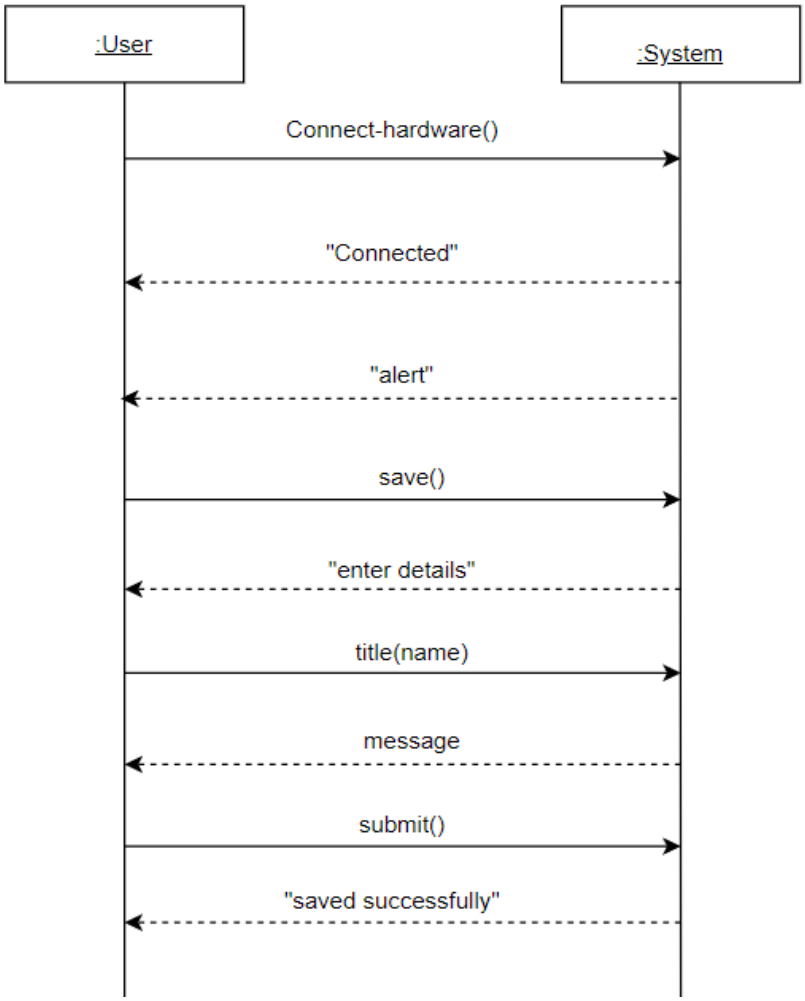


:

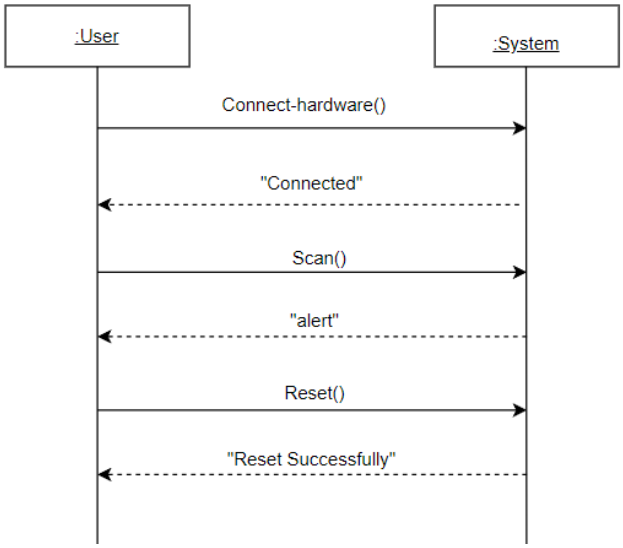
Logout:



Save Metal detection Info:

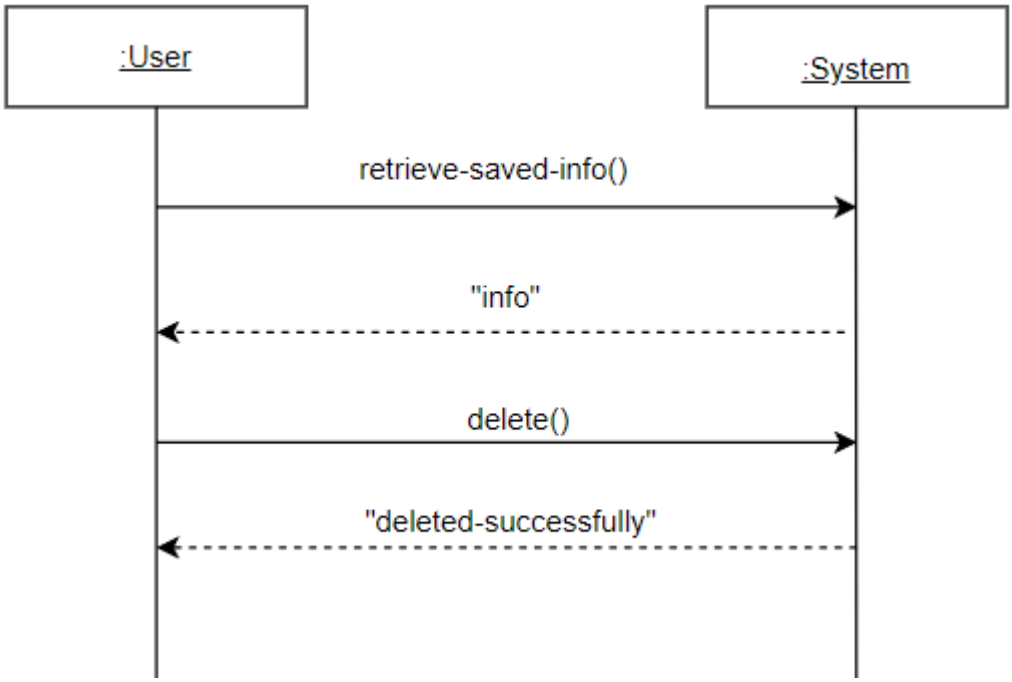


Disconnect to Metal Detector device(Reset):

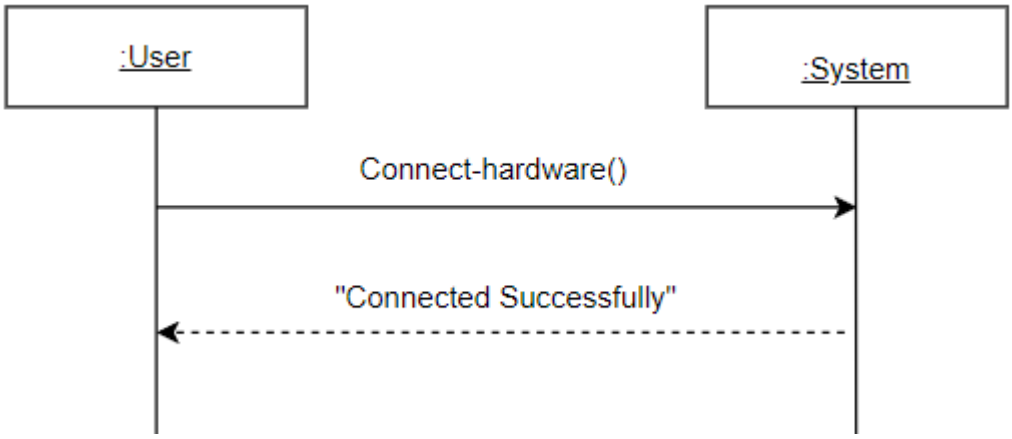


:

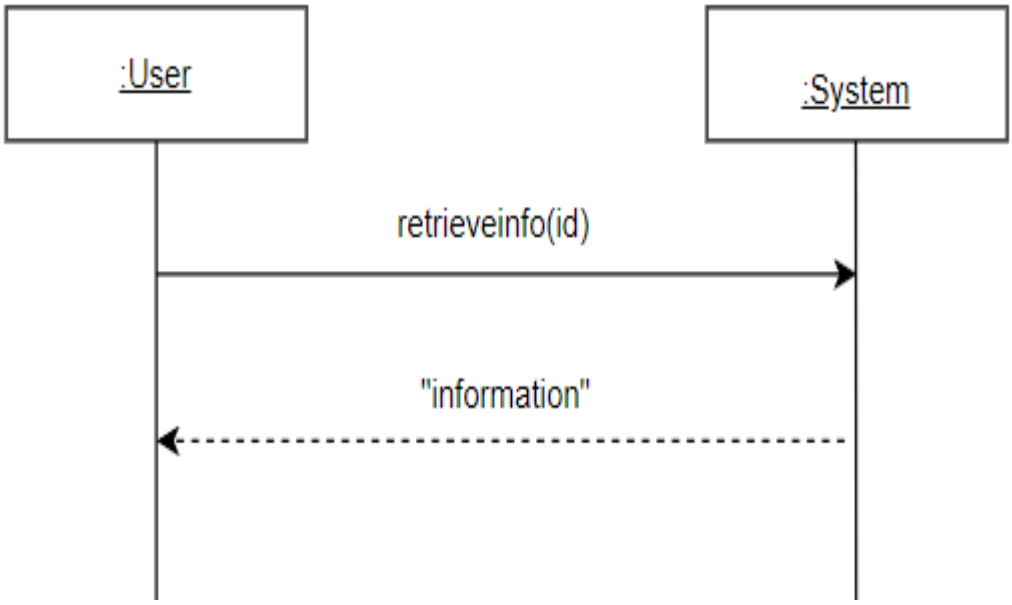
Delete metal detection Info:



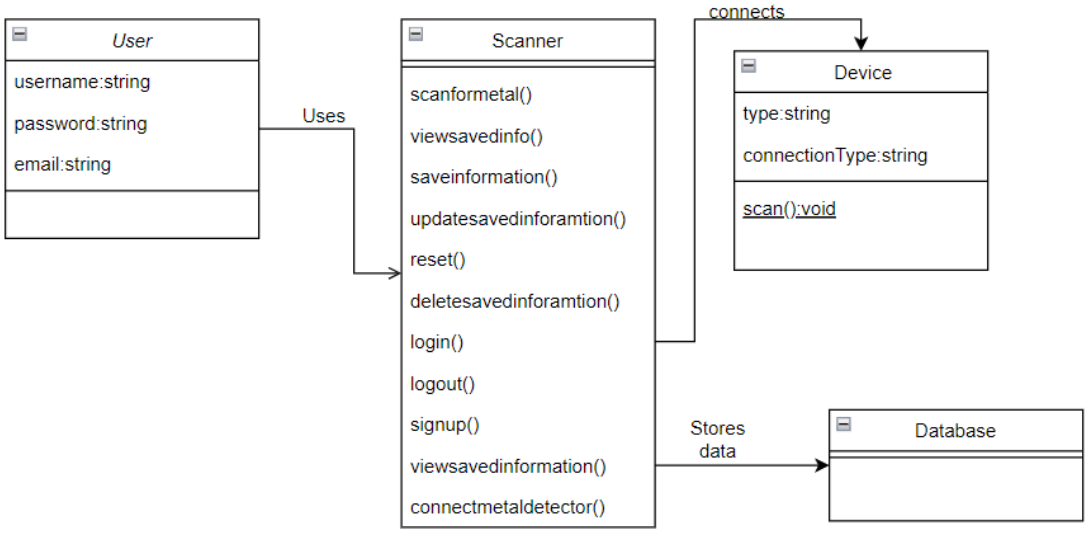
Connect Metal detector device:



View saved metal detection Info:



2.9 Domain Model:



In domain model, we represent real world classes which are related to project. The domain model of the "Metal Detector Mobile Application" is a conceptual representation that defines the fundamental entities and their relationships within the application's scope.

Chapter 3
Software Design Description

Introduction

This chapter presents a comprehensive overview of the Metal Detector Mobile Application, covering aspects such as the architecture diagram, interface designs, software structure, and its alignment with specified requirements.

3.1 Design Overview

The design phase is very important in Software development.
Thus the design of Mobile based Meta Detector Scanner helps us to handle integration with external metal detectors.
The application mainly focused on connecting metal detectors through wired interfaces.
As raw data is in analogy form generated by detectors , the main aim is to convert it into a digital format and process it.
The application will first understand and interpret that raw data and The analysis and store relevant information by converting into digital signals efficiently by employing Firebase Real time database for data management.

3.2 Requirement Traceability Matrix

As our main focus is on Signal processing capabilities so subsequent section will provide derailed components and interfaces to empower Mobile based Metal Detector Scanner Application to convert, interpret and process analog signals.
The Requirement Traceability Matrix will serve as a connection between system requirements,design components and test cases.
It will prompt the necessary actions and outcomes within the system.

Requirement ID	Requirement Name	Interface	Test Case
R1	Sign-up	User Sign-up Interface	Yes
R2	Login	User enters login credential	Yes
R3	Metal Detector Connection	Connection Interface	Yes
R4	Scan	Scanning For metal	Yes

R5	Access Saved Information	Saved data	Yes
R6	Save information	Save the data	Yes
R7	Disconnect to device	Disconnect Button	Yes
R8	Update information	Update saved data	Yes
R9	Delete data	Delete data	Yes
R10	logout	User logout interface	Yes

3.3 SYSTEM ARCHITECTURAL DESIGN

3.3.1 Chosen System Architecture

The System Architecture is very important for any software development. Thus, for my Mobile-based Metal Detector Scanner Application, I go with client model 3 Tier Architecture.

Client Tier:
The Client tier will be responsible for applications running on users' devices. It will allow users to perform actions such as registration, authentication, metal detection, and information management and will handle the user interface and user interactions.

Application Tier:
The application tier also knows ad the processing layer will be responsible for handling business logic and application functionalities. It consists of components for managing stored signal data, connecting an external metal detector, authenticating users, and receiving real-time metal detection alerts.

Data Tier:
The data tier will be responsible for storing and managing application data. A lightweight database or storage solution is utilized that will store user profiles, authentication data, and information for metal detection events.

3.3.2 Discussion of Alternative Designs

There are many architectural designs including client-server models, database options, and communication protocol but our three-tier architecture was selected for its scalability, maintainability, and separation of concerns.

3.3.3 System Interface Description

User Registration Interface:
Providing necessary information to allow users to register.

User Authentication Interface:
By entering the correct credentials. users to log in securely.

Metal Detector Connection Interface:
Makes it easier for the mobile application to connect via wired to an external metal detector.

Real-time Metal Detection Alerts Interface:
When metal objects are detected, real-time alerts are send to users.

Access to Saved Information Interface:
Saved metal detection information can be accessed by Users.

Save Processed Signal Interface:
It will allow user to processed information from metal detection events.

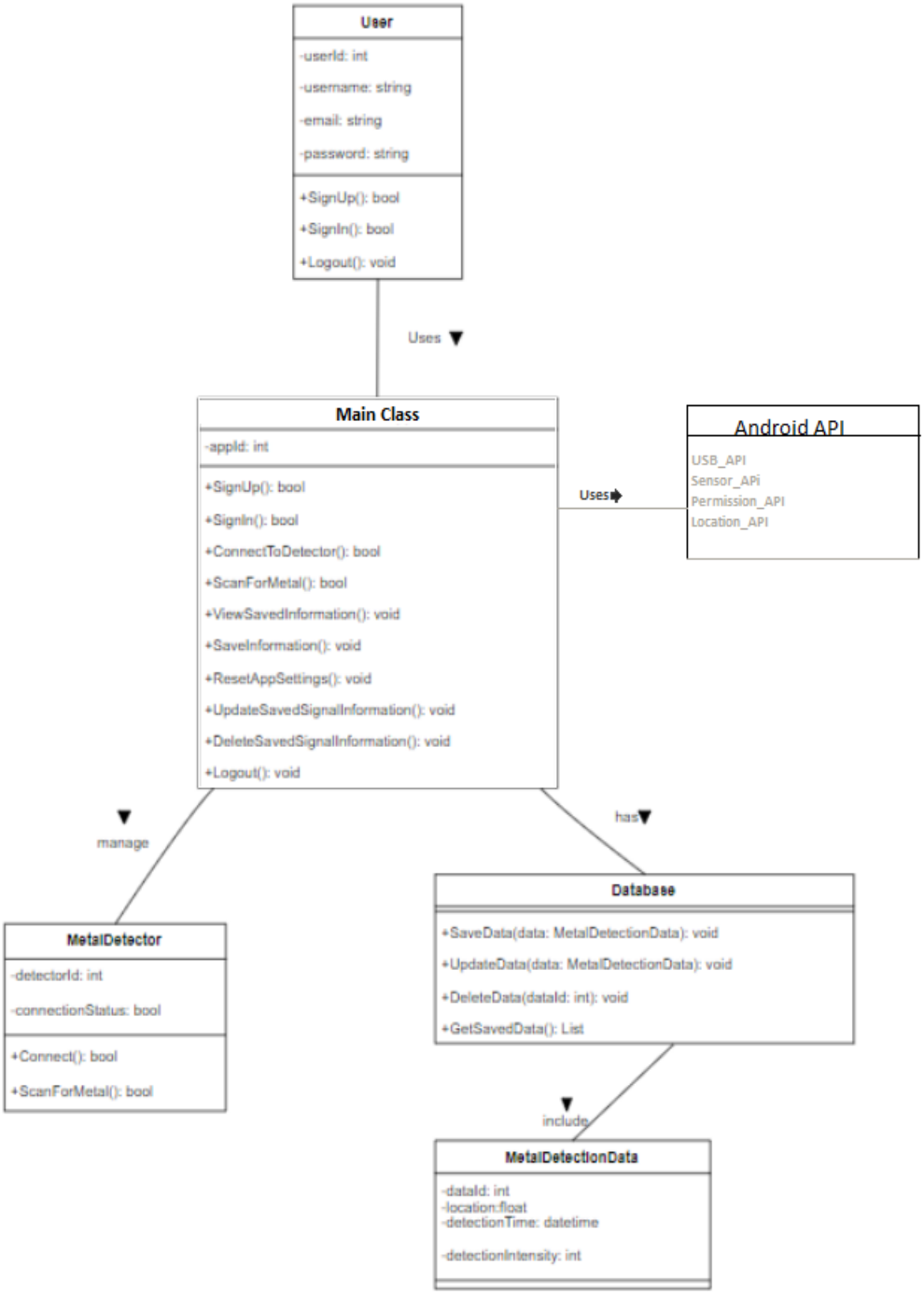
Disconnect to device Connected Interface:
Interface for user to reset the application settings and disconnect with the connected device.

Delete Saved Detection Interface:
Users can delete unnecessary saved detection entire through this interface

User Logout Interface:
To securely log out of their accounts.

As this three tier Architecture allows us to make modular systems with potential future enhancement including integration with other databases or mobile devices. It also focused on Separation of concerns to promote its maintainability though complete life cycle

3.4 Class Diagram



CHAPTER 4
IMPLEMENTATION

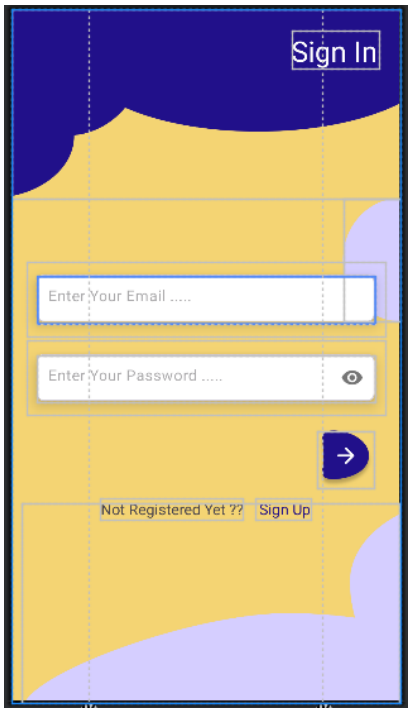
Introduction:

This chapter details the practical aspects of developing the Mobile-based Metal Detector Application. This section covers the process involved in translating the project's design into a working application. It includes the coding practices, tools, and technologies used, as well as the integration of various components. The goal is to provide a clear understanding of how each feature and functionality was realized, ensuring that the application meets the specified requirements and performs as intended.

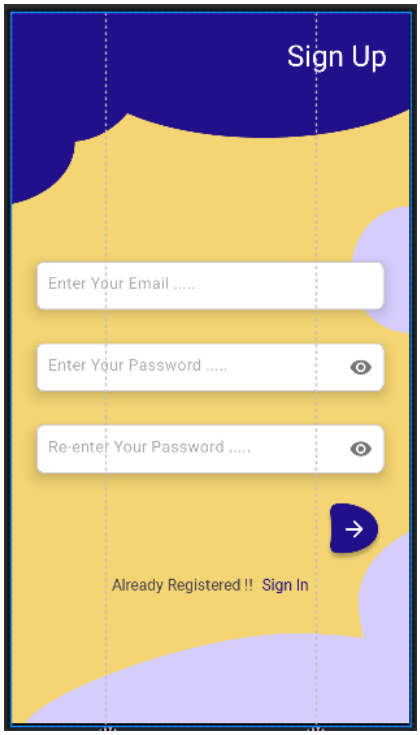
4.1 UI/UX

Splash Screen:

SignIn Screen:

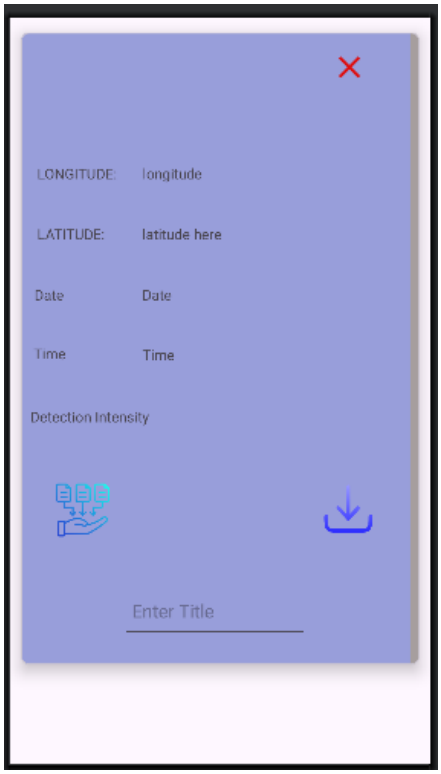


Sign Up Screen:

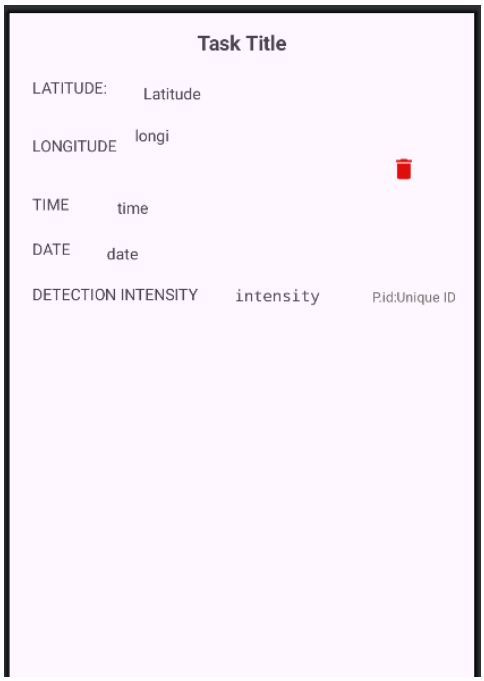


Home Screen:

Metal Detection Data screen:



Menu Screen :



4.2 Imported Libraries

The following libraries have been imported in the project to add specific functionalities beyond the default setup:

1. Google Play Services:
  - *`com.google.android.gms:play-services-location:19.0.1`*
    - Provides location services API.

2. Firebase Libraries:
  - *`com.google.firebase:firebase-database`*
    - Enables integration with Firebase Realtime Database.
  - *`com.google.firebase:firebase-auth`*
    - Facilitates user authentication with Firebase.
  - *`com.google.firebase:firebase-firestore-ktx`*
    - Adds Firestore database with Kotlin extensions for enhanced functionality.

3. USB Serial Library:
  - *`com.github.mik3y:usb-serial-for-android:3.7.3`*
    - Allows communication with USB serial devices.

4. Navigation Components:
  - *`androidx.navigation:navigation-fragment-ktx:2.7.7`*
    - Provides navigation framework for managing app navigation with Kotlin extensions.

- *`androidx.navigation:navigation-ui-ktx:2.7.7`*
  - Integrates navigation framework with UI components using Kotlin extensions.

5. Lifecycle Components:
  - *`androidx.lifecycle:lifecycle-livedata-ktx:2.5.1`*
    - Adds lifecycle-aware LiveData components with Kotlin extensions.
  - *`androidx.lifecycle:lifecycle-viewmodel-ktx:2.5.1`*
    - Adds lifecycle-aware ViewModel components with Kotlin extensions.

4.2.1 USB Serial Library:

For USB communication, I have used the usb-serial-for-android library (version 3.7.3). This library is particularly suitable for Android applications that need to interface with USB serial devices such as Arduinos. Here are some reasons why this library is advantageous and a comparison with other USB communication libraries:

Features	usb-serial-for-android	Aflak USB Library	Libusb	Android USB Library (felHR85)
Compatibility with Arduinos	✓	×	×	×
Ease of Use	✓	✓	×	×
Good Documentation	✓	✓	✓	✓
Active Maintenance	✓	✓	✓	✓
Device Specific Support	✓	×	×	✓
Platform Specific	✓	✓	×	✓
High Community Support	✓	×	✓	×
Low Setup Complexity	✓	✓	×	✓
Small Library Size	✓	✓	×	✓

Mik3y USB-Serial-For-Android:

- **Ease of Use**: This library is straightforward to integrate with Android projects. It comes with clear documentation and examples, making it easy for developers to get started.
- **Compatibility**: It supports a wide range of USB serial devices, making it a versatile choice for various projects.
- **Active Development**: The library is actively maintained, ensuring that it stays up-to-date with the latest Android versions and features.

- **License**: It's released under the MIT License, allowing for free use in both personal and commercial projects.
- **Data Transfer Speed**: Known for its efficient data transfer capabilities.
- **Error Handling**: The library has robust error handling mechanisms, making it reliable for critical applications.
- **Documentation and Community Support**: Comprehensive documentation and a large community make it easier to find support and resources.
- **Supported Platforms**: Specifically designed for Android, ensuring optimized performance on this platform.
- **Additional Features**: Supports various serial drivers, enhancing its flexibility and usability across different hardware setups.
- **Size of Library**: It is lightweight, adding minimal overhead to your application.
- **Flexibility**: High flexibility in terms of configuration and supported features, making it suitable for a wide range of use cases.

4.3 App Permissions:

Following Permissions are declared in Android Manifest File of the Application:

· **USB Permissions**:

- Added <uses-permission android:name="android.permission.USB_PERMISSION" /> to request permission for USB access.
- Included <uses-feature android:name="android.hardware.usb.host" /> to declare the app's use of USB host features.

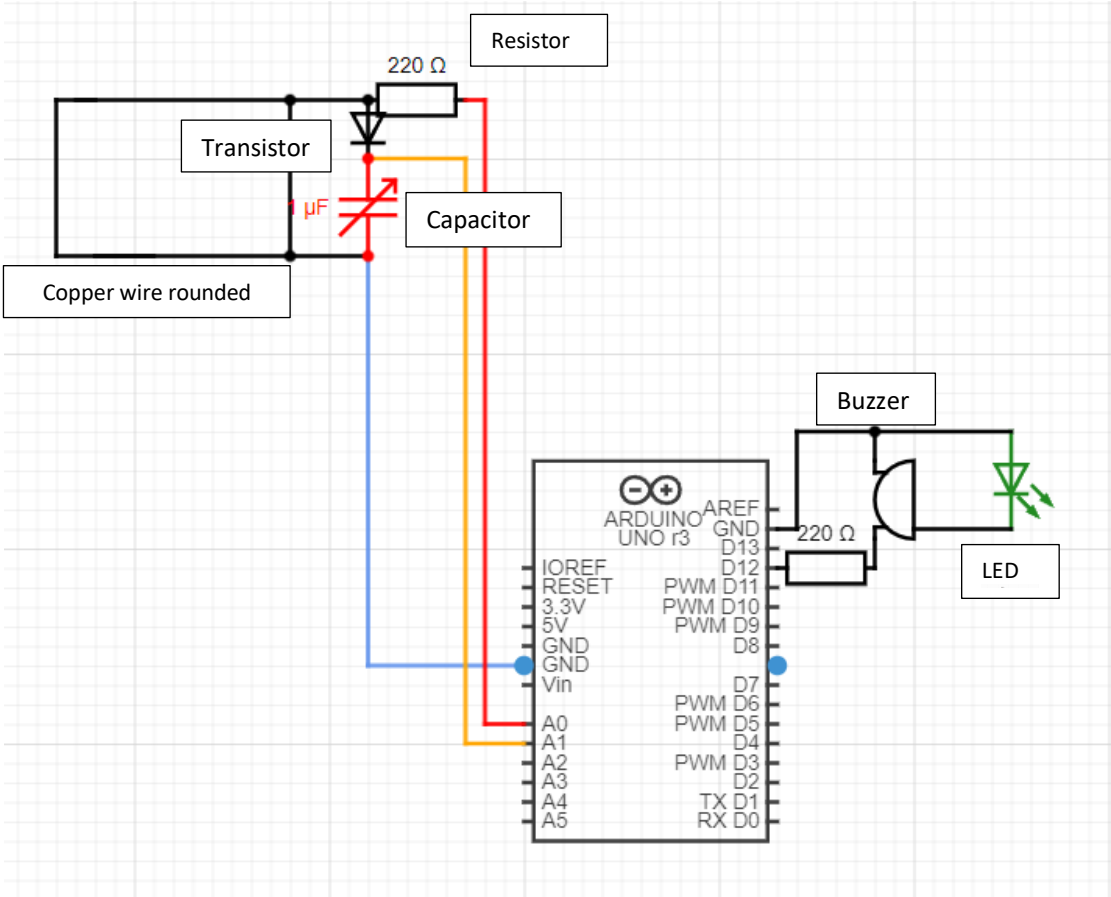
· **Location Permissions**:

- Added location permissions with <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/> and <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/> to allow access to the device's location services.

4.4 Compatibility:

- **Devices**: This Application is compatible with devices running Android 10 (API level 29) and above. Users with devices running Android 10, 11, 12, or 13 can install and run the app, while those with Android versions older than 10 will not be able to access it.
- **Behavior**: Setting the targetSdk to 33 allows the app to behave as expected on devices running Android 13 and provides a more stable experience when interacting with external hardware like Arduino.

4.5 Metal detector (External device) :



To design and implement the metal detector for my final year project, I referenced the YouTube video "How to Make a Metal Detector at Home" by Hacktuber. This video provided a comprehensive guide, detailing the components required and the assembly process. By following these instructions, I was able to create a functioning metal detector, which formed the basis of my project. The video was instrumental in understanding the practical aspects of metal detector construction, and I adapted the design to suit my project's specific requirements. You can view the tutorial here: Hacktuber's Metal Detector Video.

4.5.1 Sensor Input Configuration

Although the Arduino setup can support multiple sensor inputs, in this project, it operates with a single sensor. This focused configuration ensures precise data collection and processing for the specific application of metal detection.

4.5.2 Voltage Application and Induction Phenomenon

A 5V voltage is applied to the metal detector coil. When the coil is brought near metal, an induction phenomenon occurs, causing AC voltage to be induced in the circuit (copper wire). This induction leads to a change in voltage, which must be calibrated to accurately measure metal detection.

4.5.3 Circuit Components and Their Roles

The red wire (pin) serves as the base voltage reference, while the orange wire (pin) provides the superposed voltage. A diode is used to convert AC to DC, and a resistor is employed to manage the current to a certain threshold. The induction effect creates AC ripples, and a capacitor is used to filter these ripples, ensuring a stable output for accurate detection.

Chapter 5
Software Test Documentation.

5.1 Introduction

This documentation offers a thorough overview of the plans, and tools that are utilized in the whole project. This ensures the robustness and reliability of the Mobile-based Metal Detector Application. This part of document validates the application's functionality, potential issues and verify that the software works properly according to its requirements.

5.1.1 System Overview

This Metal Detector Application is specialized to detect the object made of metal in the real-time with its connectivity with the external metal detector. The user has to register first, then sign in into the application, connect the metal external metal detector. Then the application scans the metal, process the data and then display the result.

5.1.2 Test Approach

In order to test the Mobile Application, we follow the comprehensive and systematic methodology. I also encompass the functional and non-functional testing. This will help to observe the application's behavior under the diverse conditions. It include different types of testing like unit, system and user acceptance testing in-order to confirm the functionality of each component of the entire system.

5.2 Test Plan

The test plan for this Mobile Application outlines the objectives, scope, resources and activities that are required for the testing procedure. It defines the each testing phase, also its entry and exit point. In addition, the plan addresses the environment for the testing thoroughly. It also explains the data requirements for test, and specifies for determining the failure or success for individual test.

5.2.1 Features to be Tested

The features to be tested include:
Sign up
Sign-in
Connecting to the Metal Detector device
Metal detection on real time
Disconnect to Device
Saving information
User logging out
Each feature will undergo the testing to Ensure its functionality with the specified requirements.

5.2.2 Features Not to be Tested

Certain features, such as database-related functionalities, are not within the scope of direct testing, as the primary focus is on the application's ability to handle metal detection processes. Additionally, hardware-specific functionalities of the external metal detector fall outside the direct testing scope.

5.2.3 Testing Tools and Environment

The environment for the testing is explicitly designed to mirror the real-world scenarios, both artificial and natural scenarios in order to validate the Mobile Application's performance.

This chapter lays the groundwork for the testing process that guarantee that the application is aligned with quality and reliable standards and function in the diverse range of scenarios.

5.3 Test Cases

1. Sign up Success

ID	**T-1**
Description	Verify that a user can successfully sign up for the application.
Setup	Launch the mobile application.
Instruction	Navigate to the SignUp screen, enter valid user details, and submit the form.
Expected Results	The user should be registered successfully, and the application should navigate to the home screen.
Verdict	Pass

2. Sign In Success

ID	**T-2**
Description	Confirm that a registered user can sign in successfully.
Setup	Launch the mobile application.
Instruction	Navigate to the SignIn screen, enter valid credentials, and log in.
Expected Results	The user should be authenticated, and the application should display the main features.
Verdict	Pass

3. Connect To Metal Detector Success

ID	**T-3**
Description	Test the successful connection to an

	external metal detector.
Setup	Ensure the mobile app is installed and the external metal detector is available and in range.
Instruction	Open the app, go to the Metal Detection settings, and establish a connection to the metal detector.
Expected Results	The app should successfully connect to the metal detector without errors.
Verdict	Pass

4. Scan For Metal Success

ID	**T-4**
Description	Validate the app's ability to scan for metal using the connected detector.
Setup	Ensure the mobile app is connected to the metal detector.
Instruction	Trigger the metal detection feature and observe the results.
Expected Results	The app should accurately detect and display information about the detected metal object.
Verdict	Pass

5. View Information Success

ID	**T-5**
Description	Check if the app can successfully display saved metal detection information.
Setup	Have previously saved metal detection data.
Instruction	Navigate to the View Information screen.
Expected Results	The app should display a list of saved metal detection information.
Verdict	Pass

6. Save Information Success

ID	**T-6**
Description	Verify that users can save metal detection information successfully.
Setup	Have the metal detection feature triggered and display information.
Instruction	Save the displayed metal detection information.

Expected Results	The information should be saved and accessible in the View Information section.
Verdict	Pass

7. Disconnect to Device

ID	**T-7**
Description	Test the reset functionality.
Setup	Have some settings or data that can be reset.
Instruction	Trigger the reset functionality.
Expected Results	The app should return to its initial state, and relevant data/settings should be cleared.
Verdict	Pass

8. Delete Saved Signal Information Success

ID	**T-8**
Description	Confirm the ability to delete saved metal detection information.
Setup	Have some saved metal detection data.
Instruction	Select a saved entry and delete it.
Expected Results	The selected entry should be removed from the saved information list.
Verdict	Pass

9. Logout Success

ID	**T-9**
Description	Test the user logout functionality.
Setup	Ensure the user is logged in.
Instruction	Trigger the logout functionality.
Expected Results	The user should be logged out, and the application should return to the login screen.
Verdict	Pass

5.4 Summary

The software testing chapter is discussion about the test strategies and plans that will be used for metal detector application.The testing approach include both functional and non functional aspects through unit testing , integration testing and user acceptance testing.
Main features of of test plan include user registration , sign in , metal detector connection, real time metal detection and signal information management
The chapter focuses on the specific areas such as database-related functionalities and hardware -specific aspects.
Testing tools aims to ensure the application is robust. This chapter present the framework for testing and ensure that application is according to requirements and reliable in performance.

CHAPTER 6
CONCLUSION

6.1 Conclusion

In summary , this Mobile Application has been a important undertaking which mixes mobile technology with the external device.The project's main goal is to Understand the USB Communication of the mobile with the External Analogue Device.Additionally it aims to furnish users with an easily navigable and user friendly interface for managing metal detection activities, capturing essential data such as location, time, and metal detecting intensity. This application increases the efficacy of metal detection processes as it is positioned at the crossroads of mobility and utility.

Starting with a clear understanding of objectives mentioned in the Research Project Accounts Documents Tracking System, in project we used technologies like Android Studio, the Kotlin Programming language, and the Firebase Real time Database. Successfully delivering key features mentioned in the project overview, such as real-time metal detection alerts and signal information management.

The systematic software development approach, as we mentioned in the Software Process Model, allowed for continual refinement based on evolving requirements and user’s feedback . The project's well organized structure, defined roles and responsibilities, facilitated seamless flow of the project.

This documentation covers almost every details of the project such as project organization, Use cases, system design, and testing strategies. The system design is represented through use case diagrams, sequence diagrams, and class diagrams, which provide a comprehensive view of Application's Functionality and architecture. The software testing documentation highlights the plan, approach , and tools used to validate the application's robustness. The key features of the application have undergone thorough examination, ensuring reliability and user-friendliness, through methodical testings.

In short, the Mobile-based Metal Detector Scanner Application represents a successful hardware and software fusion, meeting the needs of users requiring correct metal detection capabilities. The project has fulfilled its aim, by delivering a functional and thoroughly tested application in alignment with the initial project goals. For Mobile based Applications, this journey represents the commitment to quality and innovation.

6.2 References:

i. IEEE Std 830-1998 (Revision of IEEE Std 830-1993), IEEE Recommended Practice for Software Requirements Specifications.
ii. Software Engineering, a Practitioners Approach by Rogers Pressman, 7th Edition.
iii. IEEE recommended practice for software requirements specifications (Approved 25 June 1998, IEEE SA Standards Board).
iv. Android Studio documentation: https://developer.android.com/docs